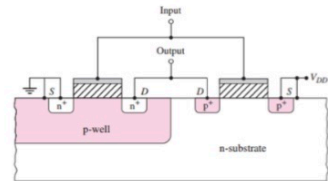
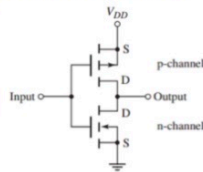
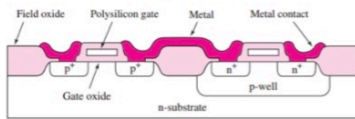
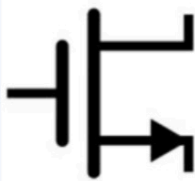
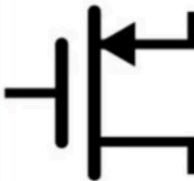


CMOS Logic Families 2

CMOS Structure



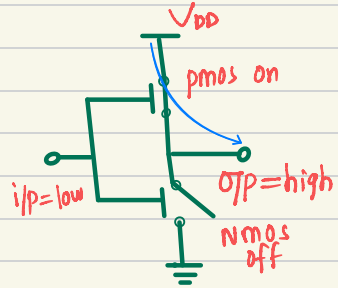
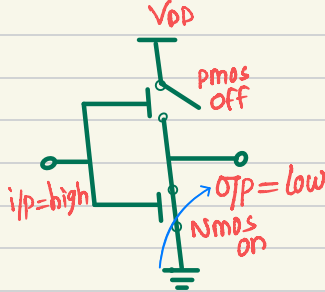
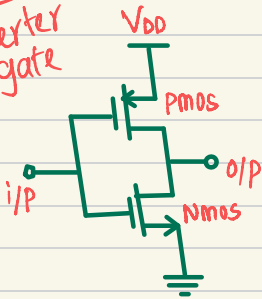
Summary:

	NMOS	PMOS
High Voltage at Gate	ON	OFF
Low Voltage at Gate	OFF	ON
Symbol	 Outward arrow	 Inward arrow

Logic Circuits using CMOS

- ⊗ If the gate voltage is high/Low the NMOS will turn on/off and PMOS will turn off/on. If any MOSFET turns on then the drain and source terminal would be short circuited.

Example:
CMOS Inverter
or, Not gate



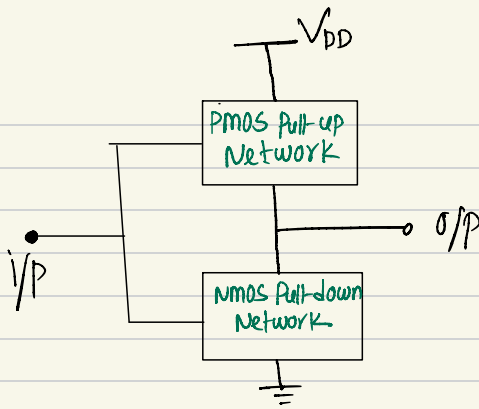
⊗ Conduction complemented

We will design "pull-up network" using PMOS & "pull down" network using NMOS

- Conduction complement means if NMOS are in series/parallel connection then PMOS are going to be in parallel/series.
- Pull-up network rises the o/p level to the logical high voltage.
- Pull-down network declines the o/p level to the logical low voltage.

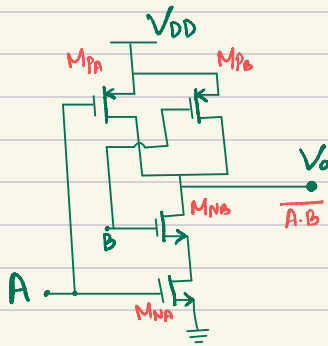
- ⊗ In order to design pull-down network we must use series NMOS connection for AND logic and parallel NMOS connection for OR logic under the whole inverse sign.

- Design NMOS pull-down network first using the logic function under the inverse sign, then design the PMOS pull-up network just by complementing the NMOS network.



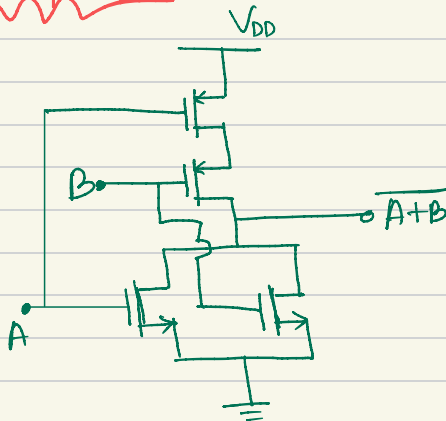
CMOS Logic Circuit

Example: CMOS NAND Gate:

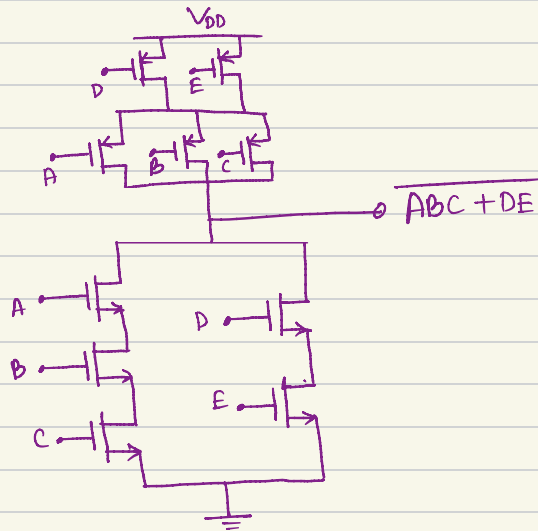


A	B	M_{PA}	M_{PB}	M_{NA}	M_{NB}	$V_0 = \overline{A.B}$
0	0	ON	ON	off	off	1
0	1	ON	off	off	ON	1
1	0	off	ON	ON	off	1
1	1	off	off	ON	ON	0

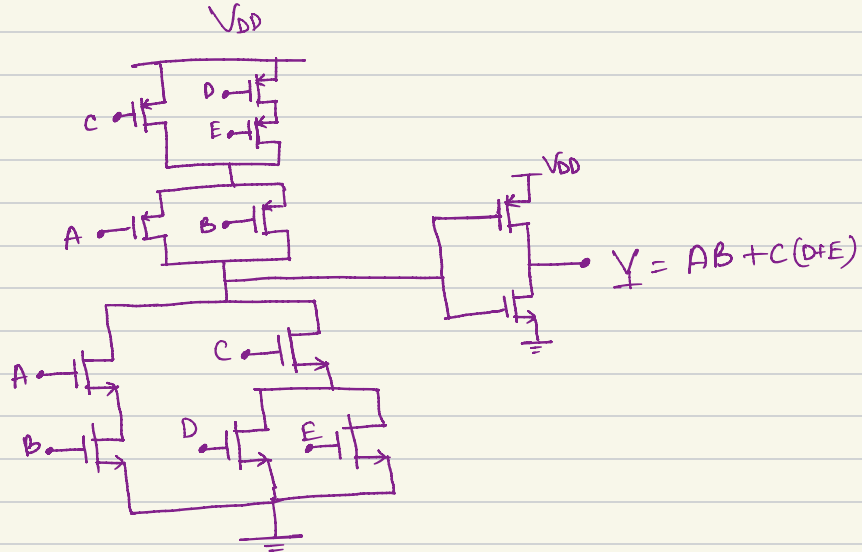
Example: CMOS NOR Gate:



Example: $Y = \overline{ABC + DE}$

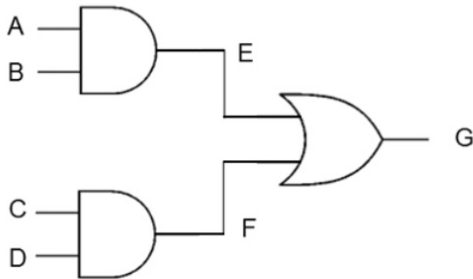


Example: $Y = AB + C(D+E)$



CMOS logic design

- Design a CMOS logic circuit to implement the given compound gate in Figure below. First derive the logical expression of output Y and then design the CMOS network.



- Design a static CMOS logic circuit that implements the logic function $Y = AB$
 - Design a static CMOS logic circuit that implements the logic function $Y = (A+B)$
- Design a static CMOS logic circuit that will implement the following logic
 - NAND gate ($Y = \overline{AB}$)
 - XOR gate ($Y = \overline{AB} + \overline{A}B$)
- Design a static CMOS logic circuit that will implement the following logic
 - NOR gate ($Y = \overline{A + B}$)
 - XNOR gate ($Y = AB + \overline{A}B$)
- Design static CMOS circuit for the following expression,
 - $Y = AB + CD$

- $Y = AB + C$
- $Y = (A+B)C$
- $Y = (A+B)(C+D)$
- $Y = \overline{AB} + \overline{CD}$
- $Y = \overline{AB} + \overline{C}$
- $Y = \overline{(A + B)C}$
- $Y = \overline{(A + B)(C + D)}$
- $Y = \overline{A} + \overline{B} + \overline{C}$

5.

Truth Table

Input A	Input B	Output
0	0	1
0	1	0
1	0	0
1	1	1

Design a static CMOS logic circuit that will implement the above truth table.